Advanced Astronomy: Solar System Standards

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AA: Solar System

This draft follows the website packet sequence, but the standards below have been updated primarily from the earlier course-draft VLO document rather than inferred packet-by-packet. Shared geometry material remains coded as `aa.x.*` because it belongs to the common AA:X strand even when it is used early in AA:SS.

Geometry (shared AA:X prerequisite)

Topic	I am able to…	Code (aa.x.euclidean_geometry.)
Triangles	… use sines, cosines, tangents, and Pythagoras' Theorem to solve right triangles quantitatively.	triangles.010
	… identify which geometric tool is appropriate in a given applied problem.	triangles.020
	… apply right-triangle geometry to real astronomical estimation problems involving lengths, distances, and angles.	triangles.030
Radians	… explain what a radian measures geometrically.	radians.010
	… use radians to determine arc lengths and related quantities.	radians.020
Units	… convert between degrees and radians.	units.010
	… define parsecs (pc) and astronomical units (au), and convert among those units and meters.	units.020

Celestial Sphere, Prominent Stars, Constellations

Topic	I am able to…	Code (aa.ss.celestial_sphere.)
Diurnal motion / rotation / revolution	… explain what diurnal motion is.	diurnal_motion.010
	… clearly distinguish rotation from revolution.	diurnal_motion.020
	… explain what prograde and retrograde motion are.	diurnal_motion.030
	… explain that the orbital motion of the major Solar System bodies is generally prograde.	diurnal_motion.040

Topic	I am able to…	Code (`aa.ss.celestial_sphere.`)
	… explain that the rotational motion of most major Solar System bodies is generally prograde.	`diurnal_motion.050`
	… use my understanding of these motions to predict the phase and sky position of the Moon from given observational information.	`diurnal_motion.060`
	… explain what it means for the Moon to be tidally locked to the Earth, and relate this to its rotation and revolution.	`diurnal_motion.070`
Ecliptic / equator / seasons	… explain what the ecliptic and ecliptic plane are.	`ecliptic_seasons.010`
	… explain what the Earth's equatorial plane is.	`ecliptic_seasons.020`
	… relate seasonal observations to the inclination of the Earth's rotation axis.	`ecliptic_seasons.030`
	… explain why summers are warm and winters are cold.	`ecliptic_seasons.040`
	… explain what the equinoxes and solstices are, when they occur, and where the Earth is in its orbit when they occur.	`ecliptic_seasons.050`
Eclipses	… give geometric explanations of solar and lunar eclipses.	`eclipses.010`
	… explain what is happening physically when solar and lunar eclipses occur.	`eclipses.020`
	… explain when eclipses can occur and when they cannot.	`eclipses.030`
	… explain the phases of the Moon during eclipses.	`eclipses.040`
	… determine the maximum lengths of eclipses.	`eclipses.050`
Celestial coordinates	… explain position on Earth in terms of latitude and longitude.	`coordinates.010`

Topic	I am able to…	Code (aa.ss.celestial_sphere.)
	… explain the horizontal coordinate system and relate it to the horizon, latitude, and altitude.	`coordinates.020`
	… locate specific objects in the sky using an appropriate coordinate system.	`coordinates.030`
Astranatomy	… explain the angular sizes subtended by common hand configurations used for rough observation.	`astranatomy.010`
	… estimate angular sizes of objects and angular distances on the celestial sphere.	`astranatomy.020`
Precession / nutation	… explain what precession and nutation are.	`precession_nutation.010`
	… explain the influence of the Moon on the Earth's precession.	`precession_nutation.020`

Aristarchus of Samos

Topic	I am able to…	Code (aa.ss.aristarchus.)
History	… explain who Aristarchus was and what he believed about the Sun, planets, and stars.	`history.010`
	… explain some of the reasons Aristarchus' model was rejected by his contemporaries, including arguments involving parallax, wind, and the motion of objects on Earth.	`history.020`
Sun–Earth–Moon	… apply Aristarchus' logic to modern observations to determine relative sizes of the Sun, Earth, and Moon.	`sun_earth_moon.010`

Topic	I am able to…	Code (`aa.ss.aristarchus.`)
	… apply Aristarchus' logic to modern observations to determine relative sizes of the Sun–Earth and Earth–Moon distances.	`sun_earth_moon.020`
	… apply Eratosthenes' method to modern observations to determine the size of the Earth.	`sun_earth_moon.030`

Kepler's Laws

Topic	I am able to…	Code (`aa.ss.kepler.`)
Copernicus	… explain that Copernicus openly argued that the Earth and other known planets orbit the Sun.	`copernicus.010`
	… explain why the Copernican model did not automatically outperform the Ptolemaic model while both still used circular orbits.	`copernicus.020`
	… state that Copernicus calculated the orbital periods of the planets.	`copernicus.030`
Tycho / Kepler	… explain who Tycho Brahe and Johannes Kepler were.	`tycho_kepler.010`
	… give a rough explanation of how Kepler used Tycho's data and Copernican ideas to obtain his first two laws.	`tycho_kepler.020`
	… explain Kepler's First Law and how it corrected the chief geometric flaw in the Copernican model.	`tycho_kepler.030`
	… explain Kepler's Second Law.	`tycho_kepler.040`
Ellipses	… construct ellipses geometrically.	`ellipses.010`

Topic	I am able to…	Code (aa.ss.kepler.)
	… represent ellipses algebraically in Cartesian and polar coordinates.	ellipses.020
	… explain and relate the ideas of semi-major axis, semi-minor axis, eccentricity, periapsis, and apoapsis.	ellipses.030
Applications of Kepler's Laws	… state Kepler's First Law.	applications.010
	… explain Kepler's Second Law in terms of motion on an ellipse and relate it to a conservation law, namely angular momentum.	applications.020
	… explain what Kepler's Second Law means for orbital speed near perihelion and aphelion.	applications.030
	… apply Kepler's Third Law to predict orbital periods or orbital radii from a table of known values.	applications.040
	… generalize Kepler's Third Law to other systems, such as the Galilean moons, and explain what changes and why.	applications.050

Solar System I

Topic	I am able to…	Code (aa.ss.solar_system1.)
Features / structure	… describe the gross features of the Solar System in terms of the Sun, the four terrestrial planets, the four giant planets, the main asteroid belt, the Kuiper belt, and the Oort cloud.	features_structure.010

Topic	I am able to…	Code (`aa.ss.solar_system1.`)
	… explain the relative sizes of the major Solar System bodies.	`features_structure.020`
	… explain the relative distances between the major Solar System bodies.	`features_structure.030`

Solar System II

Topic	I am able to…	Code (`aa.ss.solar_system2.`)
Formation theory	… explain nebular theory in terms of interstellar matter (ISM), giant molecular clouds (GMCs), dust, radiative cooling, Jeans instability, cloud collapse and fragmentation, angular momentum, spin-up, flattening, frictional heating, the T Tauri phase, and gas expulsion.	`formation_theory.010`
	… explain qualitatively what angular momentum is and how its conservation leads to spin-up and flattening of the protoplanetary disk.	`formation_theory.020`
	… explain what accretion is and how planetesimals form.	`formation_theory.030`
	… explain how heating of the collapsing protostellar bulge contributed to planetary differentiation.	`formation_theory.040`
	… explain what the asteroid belt is and why it exists.	`formation_theory.050`
	… explain the roles of the gas giants in the formation of the Kuiper belt and Oort cloud.	`formation_theory.060`

Topic	I am able to…	Code (aa.ss.solar_system2.)
	… describe with proper vocabulary what is happening in numerical simulations of GMC collapse and fragmentation.	formation_theory.070
	… describe the temperature profile of the proto-solar system, identify the ice line, describe the formation of refractory materials, and use this to explain planetary differentiation.	formation_theory.080
	… describe gravitational assists by appealing to conservation of linear momentum, and relate them both to space travel and to Solar System structure.	formation_theory.090

Solar System III

Topic	I am able to…	Code (aa.ss.solar_system3.)
Quantitatively	… apply the Virial Theorem to determine conditions for the stability of a warm, self-gravitating molecular cloud and the onset of Jeans instability.	quantitative.010
	… calculate the Jeans mass M_J or critical temperature T_c from thermodynamic stability arguments for a cloud fragment.	quantitative.020
	… calculate the free-fall time $t_{\rm ff}$ for the collapse of a giant molecular cloud from its initial density.	quantitative.030

Topic	I am able to…	Code (aa.ss.solar_system3.)
	… calculate how the temperature of the central region increases as a function of the size of a collapsing cloud.	`quantitative.040`
	… construct a realistic model of the temperature profile of the proto-solar system and identify the ice line.	`quantitative.050`
	… calculate the change in velocity, Δv, of a small body in a one-dimensional gravitational assist.	`quantitative.060`

Sol

Topic	I am able to…	Code (aa.ss.sol.)
Structure	… describe the Sun in terms of its total mass, radius, density, luminosity, and general observational character.	`structure.010`
	… describe the structure of the Sun in terms of the core, radiation zone, convection zone, photosphere, chromosphere, and corona.	`structure.020`
	… describe the Sun's temperature structure and differential rotation.	`structure.030`
	… explain hydrostatic equilibrium and zonal flow in the context of the Sun.	`structure.040`
Solar wind / weather	… describe the basics of radiation using the Planck curve, Wien's Law, and the Stefan–Boltzmann law.	`solar_wind_weather.010`

Topic	I am able to…	Code (`aa.ss.sol.`)
	… calculate the Sun's surface gravity, escape speed, luminosity, and surface temperature.	`solar_wind_weather.020`
	… use those physical quantities to reason about solar mass loss.	`solar_wind_weather.030`
	… describe the Sun's magnetic field and its implications for polarity, sunspots, and the solar cycle.	`solar_wind_weather.040`
	… describe solar flares, coronal mass ejections (CMEs), and space weather, including the hazards they pose for Earth.	`solar_wind_weather.050`
Fusion	… describe how energy is generated in the core using basic nuclear particle physics, the mass–energy relation, and the proton–proton chain.	`fusion.010`
	… identify the roles of protons, neutrons, electrons, neutrinos, and nuclei in solar fusion processes.	`fusion.020`
	… calculate the rate of solar energy generation in an appropriate simplified model.	`fusion.030`

Jupiter I

Topic	I am able to…	Code (`aa.ss.jupiter1.`)
Bulk properties / rotation	… describe Jupiter in terms of its total mass, size relative to Earth and the Sun, average density, and the implications of that density for composition.	`bulk_properties_rotation.010`

Topic	I am able to…	Code (`aa.ss.jupiter1.`)
	… describe Jupiter's rapid rotation rate and its major observational consequences.	`bulk_properties_rotation.020`
	… estimate and compare Jupiter's surface gravity and equatorial rotation speed using order-of-magnitude and scaling arguments.	`bulk_properties_rotation.030`
Rotating frames	… describe the Coriolis pseudo-force and how it gives rise to zonal flow.	`rotating_frames.010`
	… describe the centrifugal pseudo-force and how it relates to flattening and oblateness.	`rotating_frames.020`
	… use the right-hand rule to determine the directions of relevant pseudo-forces in a rotating frame.	`rotating_frames.030`
Atmosphere / weather	… describe the gross features of Jupiter's atmosphere in terms of bright zones, dark belts, large-scale storms, east–west zonal flows, and differential rotation.	`atmosphere_weather.010`
	… explain how rapid rotation and pseudo-forces organize atmospheric motion and lead to alternating bands with opposite flow directions.	`atmosphere_weather.020`
	… identify the role of the Coriolis pseudo-force in shaping Jupiter's banded structure and large-scale atmospheric circulation.	`atmosphere_weather.030`
Interior / composition	… describe what is and is not understood about the interior structure of Jupiter.	`interior_structure.010`

Topic	I am able to…	Code (`aa.ss.jupiter1.`)
	… describe Jupiter's internal structure qualitatively, including outer gaseous layers, deeper fluid regions, and a dilute or uncertain core.	`interior_structure.020`
	… describe what a phase diagram represents and use the phases of hydrogen to draw conclusions about Jupiter's interior.	`interior_structure.030`
	… explain why hydrogen is expected to behave like a metal at depth in Jupiter because of the extremely high pressures there.	`interior_structure.040`
	… explain why Jupiter radiates more energy than it receives from the Sun, and relate this to Kelvin–Helmholtz contraction and the slow release of gravitational energy.	`interior_structure.050`
Magnetosphere	… describe the origin of Jupiter's strong magnetic field in terms of rapid rotation and electrically conducting material in the deep interior.	`magnetosphere.010`
	… describe the scale and significance of Jupiter's magnetosphere, including trapped charged particles, radio emission, and the radiation environment near the moons.	`magnetosphere.020`

Jupiter II

Topic	I am able to…	Code (`aa.ss.jupiter2.`)
Moons and rings	… identify the Galilean moons Io, Europa, Ganymede, and Callisto.	`moons_rings.010`
	… explain what orbital resonance is and how it applies to the Galilean moons.	`moons_rings.020`
	… describe the Amalthean group—Metis, Adrastea, Amalthea, and Thebe—and locate them relative to Jupiter, its rings, and the Galilean moons.	`moons_rings.030`
	… describe Jupiter's moon-and-ring system in terms of its major subdivisions and the relationships among rings, inner moons, and Galilean moons.	`moons_rings.040`
	… describe the Galilean moons as a system, including their relative sizes, densities, orbital distances, and the trend of decreasing density with increasing distance from Jupiter.	`moons_rings.050`
	… explain the analogy between the Galilean system and a "mini solar system."	`moons_rings.060`
	… describe how the moons interact with Jupiter through gravity, tides, resonance, and the magnetic-field environment.	`moons_rings.070`
Rotating frames / pseudo-forces	… describe motion in a rotating reference frame in terms of centrifugal and Coriolis pseudo-forces.	`rotating_frames_pseudo_forces.010`

Topic	I am able to…	Code (aa.ss.jupiter2.)
	… determine when pseudo-forces must be included and when they should not.	rotating_frames_pseudo_forces.020
	… describe and identify the five Lagrange points of a two-body system and explain how they relate to Jupiter's Trojan asteroids.	rotating_frames_pseudo_forces.030
	… analyze the stability of Lagrange points, including the distinction between unstable, absolutely stable, and dynamically stable equilibrium points.	rotating_frames_pseudo_forces.040
	… predict qualitatively the motion of objects displaced slightly from the neighborhood of a Lagrange point.	rotating_frames_pseudo_forces.050